COLLEGE SPORTS ADMINISTRATION SYSTEM

A PROJECT REPORT

Submitted by

ATCHAYA R

(2116241501026)

in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "COLLEGE SPORTS ADMINISTRATION SYSTEM" is the bonafide work of ATCHAYA R (2116241501026), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026-2027, Semester V, under my supervision.

SIGNATURE	**SIGNATURE**
DR. AYYADURAI	DR. J. JASMINE GABRIEL
HEAD OF THE DEPARTMENT	**COURSE FACULTY**
Department of AIML	PROFESSOR
Rajalakshmi Engineering College	Department of AIML
Thandalam – 602 105	Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on 21/10/2026.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, ATCHAYA R (2116241501026), hereby declare that the project report titled “COLLEGE SPORTS ADMINISTRATION SYSTEM”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report – including its data model, forms, reports, dashboard and permission profiles – was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 21/10/2026

SIGNATURE OF THE STUDENT
ATCHAYA R
2116241501026

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman S. MEGANATHAN, Chairperson THANGAM MEGANATHAN and Vice-Chairman ABHAY SHANKAR MEGANATHAN for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, DR. S. N. MURUGESAN, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank DR. AYYADURAI, Head of the Department, Department of AIML, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, DR. J. JASMINE GABRIEL, PROFESSOR, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation / Zoho Creator Education Programme for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

ATCHAYA R

ABSTRACT

Sports and games are an important part of college life, yet the way a college administers them is often informal. Events are announced by notices and messages, players register by e-mail or on paper, their age and eligibility are checked by hand, the registration status and the fees are tracked in spreadsheets, and the sports coordinator has no single place to see who has registered for which event, whose registration is pending and how much fee has been recorded. This makes it hard to monitor the registrations and to prepare the participation records that are needed at the end of the season.

To address this problem, the College Sports Administration System was developed on Zoho Creator, a low-code application platform. The application has two forms. The Sports_Event form records the event name, sport category, event date, venue, maximum participants, registration fee, event coordinator and event status. The Player_Profile form records the player name, e-mail address, mobile number, age, gender, sport category, experience level, the selected events (chosen from the Sports_Event form), registration date, registration status, player feedback and total fee. Two list reports present the records of the two forms, and a dashboard page shows the total fees, the total registration fees, the number of players and events, the count of players by registration status, quick links and charts. Three form workflows written with Deluge enforce the rules of the process: one sets the registration status of a new profile to Pending, one verifies that the player is at least 12 years old before the profile is saved, and one notifies the player when the total fee is updated.

The application was tested with 17 functional test cases on live records of five sports events and five player profiles. The results show that events and players are captured through validated fields, that the age rule is enforced and that the dashboard agrees with the reports. The observations also show that the events chosen are not filtered by sport category, that the event status does not follow the event date and that the total fee is typed by hand; filtering the lookup, a scheduled status workflow, an automatic fee calculation and an approval process are proposed as the next enhancements.

Keywords: Low-Code Development; Zoho Creator; Sports Administration; Deluge Workflows; Player Registration; Dashboard

PROJECT AT A GLANCE

Project Title	College Sports Administration System
Application Domain	Education / Sports and Event Administration
Student Name & Register No.	ATCHAYA R – 2116241501026
Zoho Creator Edition	Trial Edition
Application Name (Link Name)	College Sports Administration System (college-sports-administration-system)
Published / Live URL	https://creatorapp.zoho.in/atchaya.r.2024.aiml_rajal593/college-sports-administration-system/#Section_1
Demo Credentials (per role)	Admin
Number of Forms / Subforms	2 (Player_Profile, Sports_Event) / 0
Number of Reports (by type)	List: 2 | Kanban: 0 | Calendar: 0 | Pivot/Chart: 0 | Others: 0
Number of Pages / Dashboards	1 (Dashboard)
Number of Workflows	Form: 3 | Schedule: 0 | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	3 workflow actions; Deluge scripts documented in Section 5.5 and Appendix C
Integrations	None in version 1.0 (Zoho Calendar, Zoho Analytics and a payment gateway planned)
Roles & Profiles Defined	Default Zoho Creator permission profiles
Environments Used	Development
Date of Production Deployment	20/10/2026

TABLE OF CONTENTS

ABSTRACT....iv
PROJECT AT A GLANCE....v
LIST OF TABLES....ix
LIST OF FIGURES....x
LIST OF ABBREVIATIONS....xi
INTRODUCTION....1
1.1 Background....1
1.2 Problem Statement....1
1.3 Objectives....2
1.4 Scope of the Project....2
1.5 Justification for a Low-Code Approach....2
1.6 Organisation of the Report....3
LITERATURE SURVEY AND SYSTEM STUDY....4
2.1 Study of the Existing System....4
2.2 Limitations of the Existing System....4
2.3 Low-Code versus Traditional Development....4
2.4 Literature Survey....5
2.5 Emerging Trends in Low-Code Development....6
2.6 Proposed System....6
REQUIREMENT ANALYSIS....7
3.1 Stakeholders and User Roles....7
3.2 Functional Requirements....7
3.3 Non-Functional Requirements....8
3.4 Hardware and Software Requirements....8
3.5 Use Case Model....8
SYSTEM DESIGN....11
4.1 System Architecture....11
4.2 Data Model....11
4.3 Form Design....12
4.4 Process and Workflow Design....13
4.5 User Interface and Navigation Design....14
4.6 Access Control Design....14
IMPLEMENTATION....16
5.1 Application Setup....16
5.2 Forms and Data Capture....16
5.3 Reports....18
5.4 Pages and Dashboards....19
5.5 Deluge Scripting....21

5.6 Workflows and Automation..... 22
5.7 Integrations..... 23
5.8 User Management and Role-Based Access..... 24
5.9 AI and Analytics Features..... 24
5.10 Pro-Code Extension..... 24
5.11 Mobile and Portal Access..... 24
TESTING AND DEBUGGING..... 25
6.1 Testing Strategy..... 25
6.2 Test Cases..... 25
6.3 Debugging Deluge Scripts..... 26
6.4 Defect Log..... 27
6.5 User Acceptance Testing..... 27
DEPLOYMENT, SECURITY AND GOVERNANCE..... 28
7.1 Environments and Deployment Strategy..... 28
7.2 Performance Optimisation..... 28
7.3 Security Features..... 29
7.4 Governance and Administration..... 29
RESULTS AND DISCUSSION..... 30
8.1 Application Output..... 30
8.2 Achievement of Objectives..... 30
8.3 Comparison with the Existing System..... 31
8.4 Limitations of the Low-Code Approach..... 31
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum..... 31
CONCLUSION AND FUTURE ENHANCEMENTS..... 33
9.1 Conclusion..... 33
9.2 Future Enhancements..... 33
9.3 Learning Outcomes and Career Relevance..... 33
REFERENCES..... 34
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST..... 35
APPENDIX B – COURSE OUTCOME MAPPING..... 37
APPENDIX C – COMPLETE DELUGE SOURCE CODE..... 38
C.1 Auto-Set Registration Status – Player_Profile, Successful Form Submission (Created)..... 38
C.2 Notify on Fee Payment – Player_Profile, Update of Total Fee (Edited)..... 38
C.3 Validate Age for Registration – Player_Profile, Validations on Form Submission (Created or Edited)..... 38
APPENDIX D – USER MANUAL..... 39
D.1 Accessing the Application..... 39
D.2 Instructions for the Player..... 39
D.3 Instructions for the Sports Coordinator..... 39
D.4 Instructions for the Administrator..... 39

APPENDIX E – EVALUATION RUBRIC.....40
APPENDIX F – VIVA-VOCE RECORD SHEET.....42

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application.....5
Table 2.2: Summary of Literature Survey.....5
Table 3.1: Stakeholders and User Roles.....7
Table 3.2: Functional Requirements.....7
Table 3.3: Non-Functional Requirements.....8
Table 3.4: Hardware and Software Requirements.....8
Table 3.5: Use Case Description – Register Player Profile.....9
Table 3.6: Use Case Description – Create Sports Event.....9
Table 3.7: Use Case Description – Monitor Registrations.....10
Table 4.1: Entities and Relationships.....12
Table 4.2: Field Specification – Player_Profile Form.....12
Table 4.3: Field Specification – Sports_Event Form.....13
Table 4.4: Role–Permission Matrix.....15
Table 5.1: Application Configuration Summary.....16
Table 5.2: Reports Implemented.....18
Table 5.3: Workflow Actions and Deluge Scripts.....22
Table 5.4: Workflows Implemented.....23
Table 5.5: Integrations Planned for Future Versions (Not Yet Implemented).....23
Table 6.1: Test Cases.....25
Table 6.2: Defect Log.....27
Table 6.3: User Acceptance Feedback.....27
Table 7.1: Deployment Checklist.....28
Table 7.2: Performance Optimisation Measures.....28
Table 7.3: Security Controls Implemented.....29
Table 8.1: Scenario Walkthrough and Evidence.....30
Table 8.2: Mapping of Objectives to Outcomes.....30
Table 8.3: Existing System versus Proposed Application.....31
Table A.1: Feature Coverage Checklist.....35
Table B.1: Mapping of Course Outcomes to Report Chapters.....37
Table E.1: Evaluation Rubric for the Individual Project.....40
Table E.2: Performance Level Descriptors.....40
Table F.1: Viva-Voce Questions and Responses.....42
Table F.2: Marks Summary.....42

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing System....4
Figure 3.1: Use Case Diagram of the Application....9
Figure 4.1: System Architecture of the College Sports Administration System....11
Figure 4.2: Entity–Relationship Diagram of the College Sports Administration System....12
Figure 4.3: Workflow / Process Flow of the College Sports Administration System....14
Figure 4.4: Navigation Structure of the Application....14
Figure 5.1: Player_Profile Form – Player Details, Category, Experience Level and Selected Event…17
Figure 5.2: Player_Profile Form – Registration Status, Player Feedback, Total Fee and Submit Button 17
Figure 5.3: Sports_Event Form....18
Figure 5.4: Player_Profile Report....19
Figure 5.5: Sports_Event Report....19
Figure 5.6: Dashboard – (a) Summary Panels, Registration Status and Quick Links, (b) Charts and Player Profiles, (c) Player Profiles Table....20
Figure 5.7: Deluge Script of the Workflow Auto-Set Registration Status....21
Figure 5.8: Deluge Script of the Workflow Notify on Fee Payment....21
Figure 5.9: Deluge Script of the Workflow Validate Age for Registration....22
Figure 5.10: Form Workflows of the Player_Profile Form....23

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Sports and games are an important part of college life. A sports department organises many events during the year – tournaments, leagues, meets and championships – and every event involves a series of steps: the event is announced with its category, date, venue and fee, students register as players, the coordinator checks that each player is eligible, the registration is approved or rejected, the registration fee is collected, and the event is finally held, completed or cancelled. The sports coordinator must be able to follow every player and every event through these steps.

In many colleges this follow-up is done through notices, messages, paper registers and spreadsheets. Each player registers in a different way, the information is scattered, and nobody can say at a glance which players have registered for which events, whose registration is still pending and how much fee has been received. The effort to collect and compile the information grows with the number of players and events.

This project builds the College Sports Administration System, a single application on Zoho Creator in which the coordinator records the sports events, the players register through a player profile and choose their events, the registration status is set automatically when a profile is created, the age of a player is checked before the profile is saved, the player is notified when the total fee changes, and a dashboard gives a summary of the registrations and fees at a glance.

1.2 Problem Statement

Colleges lack a simple, central system that records sports events and the registrations of players. Because registrations and fees are scattered across notices, messages and files, it is difficult to check the eligibility of players, to know the real registration status and to prepare reports. An application is needed that captures each fact once, validates it and keeps the registration status and the fee information up to date.

- Event details (category, date, venue, number of participants and fee) are announced and recorded in different formats.
- Player details are collected on paper or in messages and are incomplete or inconsistent.
- The age and eligibility of a player are not verified, so under-age players can register.
- The registration status is updated by hand and is often outdated.
- Players are not informed when their fee is updated, and the coordinator must compile the lists and reports manually.

1.3 Objectives

- To design two forms, Sports_Event and Player_Profile, that capture the events and the players with validated fields, and to link a player to the events chosen.
- To set the registration status of a new player automatically and to validate the age of a player on submission using Deluge form workflows.
- To notify the player by e-mail when the total fee of the profile is updated.
- To provide list reports and a dashboard that give the coordinator an up-to-date view of players, events, fees and registration status.
- To test the application, record the defects found and define the improvements for the next version.

1.4 Scope of the Project

In scope. The application covers two modules – Sports Events and Player Profiles – each with a data-entry form and a list report, a dashboard page with summary panels, quick links and charts, and three form workflows written in Deluge. The Player_Profile form is linked to the Sports_Event form through the Selected Event field. Dates use the dd-MMM-yyyy format and fees are shown in currency format. The application is accessible on the web and through the responsive Zoho Creator interface.

Out of scope. An approval process for registrations, automatic updating of the event status from the event date, automatic calculation of the total fee, certificates, payment-gateway integration and a player portal are not part of version 1.0. They are discussed as future enhancements in Section 9.2.

1.5 Justification for a Low-Code Approach

Low-code development platforms provide visual environments in which data models, forms, reports and workflows are assembled by configuration, with code needed only for specialised logic [1], [2]. Research comparing low-code with model-driven engineering notes that such platforms run on the cloud, reduce manual coding and are aimed at users who are not professional programmers [3]. These characteristics match the needs of this project.

The sports administration process has a small and well-understood data model, the users are players and a sports coordinator, and the rules (setting the status, checking the age and sending a notification) are short. Zoho Creator supplies hosting, a managed database, form, report and page builders and the Deluge scripting language, so the whole application could be built without any infrastructure. The rules were written as a few lines of Deluge instead of a back-end program, and the college can change them without redeploying code.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing arrangement the sports coordinator announces an event by a notice or a message. Players reply by e-mail, by phone or on paper, and the names are noted in a spreadsheet or on a register. The age and eligibility of each player are checked by hand, the registration status and the fees are tracked by hand, and at the end of the season the coordinator compiles the participation report manually. Figure 2.1 shows this as-is process.

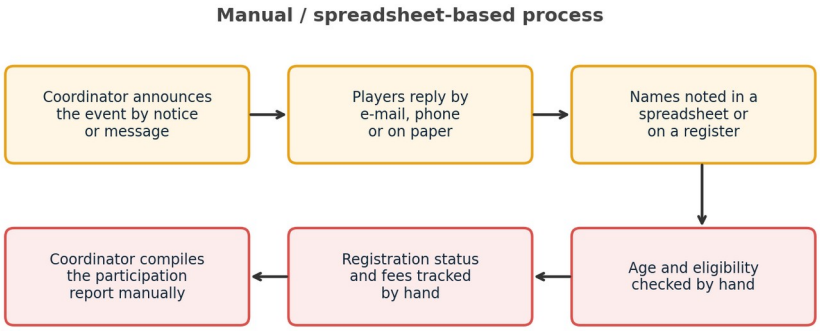


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Event and player details are collected in different formats and are incomplete or inconsistent.
- The age of a player is not verified, so an under-age player can be registered without being noticed.
- Which player has chosen which event is not recorded in one place.
- The registration status depends on someone remembering to update it.
- Players are not told when their fee changes, and anyone who holds the spreadsheet can see and change the contact details of every player.
- Reports must be compiled by hand, which delays decisions.

2.3 Low-Code versus Traditional Development

Low-code platforms are reported to shorten delivery because they automate much of the work of building and deploying business applications [4]. Table 2.1 compares the two approaches for this application. The effort figures are estimates made by the author for a comparable two-form application and are given only to illustrate the difference in order of magnitude.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 4–6 weeks for database design, back-end, front-end and authentication	Estimated 1–2 days to configure 2 forms, 2 reports, 1 dashboard and 3 workflows
Skills required	Web development, database design, server administration	Data modelling and basic platform configuration; Deluge for the rules
Hosting & infrastructure	Server, database and domain must be arranged and maintained	Hosted and managed by the platform
Mobile application	Separate mobile app or responsive front-end must be built	Available through the Zoho Creator mobile interface without extra coding
Validation & business rules	Written and tested in back-end code	Form workflows with a few lines of Deluge
Maintenance & change	Code change, testing and redeployment for every new field	A field or a rule is changed in the builder in minutes
Customisation limits	Almost unlimited	Bounded by platform features; pro-code extensions needed for complex needs
Cost	Developer time plus hosting costs	Platform subscription; no separate hosting cost

2.4 Literature Survey

Table 2.2 summarises the sources reviewed for this project: textbooks and platform documentation on Zoho Creator and Deluge, and research papers on low-code platforms.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski [1]	Low-Code Development: A Practical Guide	Explains low-code concepts, the role of the citizen developer and a step-by-step approach to building applications	Course text; guided the overall development approach
2	Zoho Corporation [2], 2026	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, permission profiles and deployment in Zoho Creator	Primary reference for configuring the application
3	D. Di Ruscio et al. [3], 2022	Low-code development and model-driven engineering: Two sides of the same coin?	Compares low-code with model-driven engineering; notes cloud execution, reduced hand coding and strengths and weaknesses	Supports the choice of a low-code platform and the discussion of its limits
4	R. Sanchis et al. [4], 2020	Low-code as enabler of digital transformation in manufacturing industry	Shows that low-code platforms automate application development and support digital transformation	Supports replacing manual tracking with a low-code application

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
5	Zoho Corporation [5], 2026	Deluge Script Reference Guide	Defines the Deluge scripting language, including conditions, operators, field assignments and tasks	Basis for the workflow scripts
6	A. Sahay et al. [6], 2020	Supporting the understanding and comparison of low-code development platforms	Presents a framework to classify and compare low-code platforms	Helps position Zoho Creator among low-code platforms
7	Zoho Corporation [7], 2026	Zoho Creator Learning Centre	Tutorials and sample applications for building Zoho Creator apps	Reference for form and workflow design practice

2.5 Emerging Trends in Low-Code Development

Several trends are shaping low-code development. AI-assisted application building allows a first version of an application to be generated from a plain-language description and then refined by hand. Citizen development lets domain experts such as a sports coordinator build and change their own tools. Hyper-automation combines workflows, scripts and integrations so that routine steps happen without manual effort, and composable applications connect small applications through APIs instead of building one large system [3], [6].

These trends influenced this project in two ways. The application was kept small and clear so that a coordinator can understand and extend it. The rules that would otherwise be done by hand – setting the registration status, checking the age and informing the player – were automated with form workflows, which is an example of hyper-automation on a small scale.

2.6 Proposed System

The proposed system is the College Sports Administration System built on Zoho Creator. It replaces notices, registers and spreadsheets with two forms, two reports and a dashboard. Each limitation listed in Section 2.2 is addressed as follows. Event and player details are captured in fixed fields with drop-downs for the sport category, gender, experience level, event status and registration status. The age of a player is verified by a validation workflow. The Selected Event field links a player to the events chosen. The registration status of a new profile is set to Pending by a workflow, and a second workflow notifies the player when the total fee is updated. The reports give the coordinator a current list of players and events at any time, and the dashboard summarises the fees, the number of players and events, and the registration status.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1 lists the stakeholders of the application and their responsibilities.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Player	Student of the college who wants to take part in sports events	Submit the player profile, choose the events and give feedback
Sports Coordinator	Faculty member or staff member in charge of sports and events	Create sports events; review registrations and set approved or rejected; update the total fee; monitor the reports and the dashboard
Administrator	Owner of the application	Manage users and permission profiles; configure forms, reports, pages and workflows

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall record a player profile with player name, e-mail address, mobile number, age, gender, sport category, experience level, selected events, registration date, registration status, player feedback and total fee.	Player, Sports Coordinator	High
FR-02	The system shall record a sports event with event name, sport category, event date, venue, maximum participants, registration fee, event coordinator and event status.	Sports Coordinator	High
FR-03	The system shall set the registration status of a newly created player profile to Pending automatically.	System	High
FR-04	The system shall validate the age of a player on submission and cancel the submission when the age is below 12 years.	System	High
FR-05	The system shall notify the player by e-mail when the total fee of the profile is updated.	System	Medium
FR-06	The system shall provide a list report for the player profiles and a list report for the sports events.	All roles	High
FR-07	The system shall allow a record to be searched, viewed, edited, duplicated or deleted from a report according to the permission profile.	Sports Coordinator, Administrator	Medium
FR-08	The system shall display dates as dd-MMM-yyyy and fees as currency with two decimal places.	All roles	Low
FR-09	The system shall provide a dashboard with the total fees, the total registration fees, the number of player profiles and sports events, the count of players by registration status, quick links and charts.	Sports Coordinator, Administrator	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 1,000 records shall load within 5 seconds on a normal broadband connection.
NFR-02	Security	Users shall access only the data permitted by their permission profile; contact details shall be protected.
NFR-03	Usability	The application shall be usable on desktop, tablet and mobile browsers and through the Zoho Creator mobile interface.
NFR-04	Availability	The application shall be available whenever the Zoho Creator cloud service is available.
NFR-05	Maintainability	A new field or rule shall be added through the builder without redeploying code.
NFR-06	Data quality	The age of a player shall be verified before a profile is saved, and dates and fees shall be stored in a fixed format.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Trial Edition at the time of development)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Other Zoho services	None in version 1.0
Mobile	Zoho Creator mobile app (Android / iOS) or a mobile browser
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases of the application for its two main actors, the Player and the Sports Coordinator. Tables 3.5 to 3.7 describe three key use cases in detail.

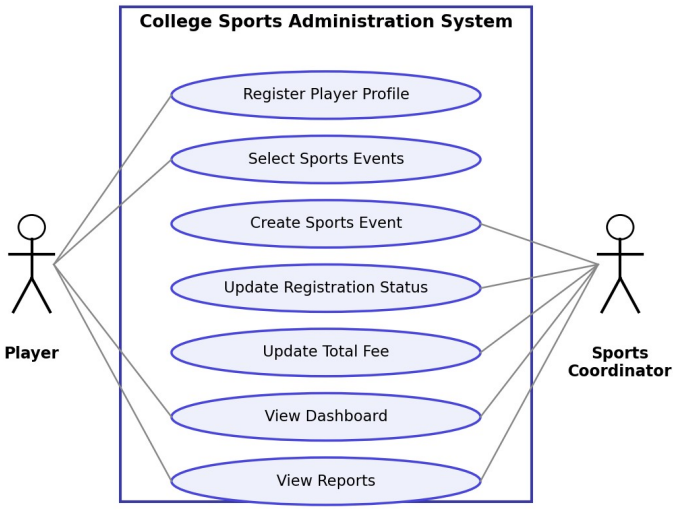


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Register Player Profile

Use Case ID	UC-01
Use Case Name	Register Player Profile
Actor(s)	Player, Sports Coordinator
Pre-condition	The user is logged in, has permission to add records to the Player_Profile form, and the sports events exist.
Main Flow	1. The user opens the Player_Profile form. 2. The user enters the player name, e-mail address, mobile number, age and gender. 3. The user selects the sport category, the experience level and the events to take part in. 4. The user enters the registration date and the total fee. 5. The user clicks Submit; the Validate Age for Registration workflow runs, the record is saved and Auto-Set Registration Status sets the status to Pending.
Alternate Flow	If the age is below 12, the system shows the message “Player must be at least 12 years old to register.” and the record is not saved.
Post-condition	The player profile appears in the Player_Profile Report with the registration status Pending.
Related FR	FR-01, FR-03, FR-04, FR-06

Table 3.6: Use Case Description – Create Sports Event

Use Case ID	UC-02
Use Case Name	Create Sports Event
Actor(s)	Sports Coordinator
Pre-condition	The user is logged in and has permission to add records to the Sports_Event form.

Main Flow	1. The user opens the Sports_Event form. 2. The user enters the event name, selects the sport category and enters the event date and venue. 3. The user enters the maximum participants and the registration fee. 4. The user enters the e-mail address of the event coordinator and selects the event status. 5. The user clicks Submit and the record is saved.
Alternate Flow	If the event needs correction, the user edits it from the Sports_Event Report and saves it again.
Post-condition	The event appears in the Sports_Event Report and can be selected in the Player_Profile form.
Related FR	FR-02, FR-06, FR-07, FR-08

Table 3.7: Use Case Description – Monitor Registrations

Use Case ID	UC-03
Use Case Name	Monitor Registrations
Actor(s)	Sports Coordinator, Administrator
Pre-condition	Player profiles and sports events exist.
Main Flow	1. The user opens the Dashboard to see the fees, the number of players and events and the registration status counts. 2. The user opens the Player Profiles report to see the players, their events and the registration status. 3. The user opens the Sports Events report to see the events, venues, fees and event status. 4. The user searches a record or opens it to read the details.
Alternate Flow	If a registration needs a decision, the user edits the profile, sets the registration status and updates the total fee; the fee update notifies the player.
Post-condition	The coordinator knows the current registration status and fees of every player.
Related FR	FR-05, FR-06, FR-07, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The solution follows the layered architecture shown in Figure 4.1. Players, sports coordinators and administrators reach the application through a desktop browser, a tablet or mobile browser, or the Zoho Creator mobile app. The Zoho Creator application layer contains two forms, two list reports, a dashboard page and three form workflows written in Deluge. All records are stored in the managed Zoho Creator cloud database, and dates are shown in the dd-MMM-yyyy format. The items shown in the data layer as planned are not part of version 1.0.

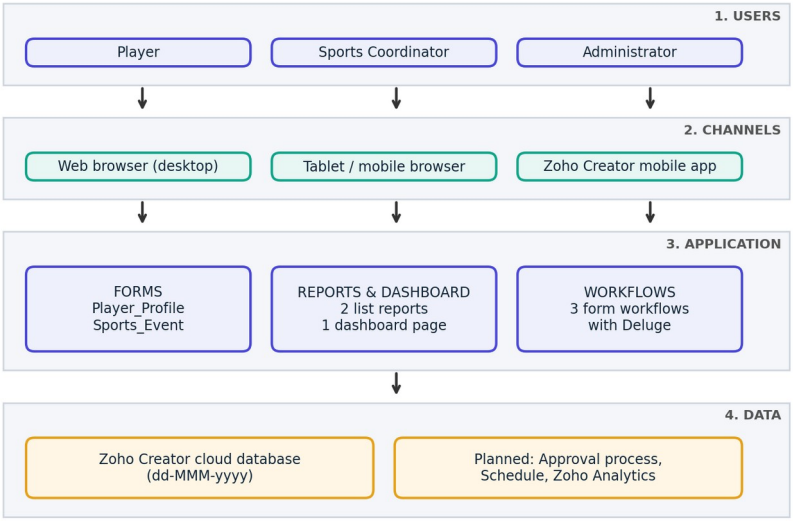


Figure 4.1: System Architecture of the College Sports Administration System

Users layer. The three roles use the application according to their permission profile. **Channel layer.** The responsive Zoho Creator interface adapts to desktop, tablet and mobile screens. **Application layer.** The forms capture data, the reports and the dashboard present it and the workflows enforce the rules. **Data layer.** The records of the two forms are stored in the platform database.

4.2 Data Model

The data model has two entities, Player_Profile and Sports_Event, shown in Figure 4.2 and Table 4.1. A sports event describes an event that the college organises, and a player profile describes a student who registers for sports. The two forms are linked through the Selected Event field of the Player_Profile form, in which the user chooses one or more events from the Sports_Event form. One player can therefore choose many events and one event can be chosen by many players. No subform is used in version 1.0.

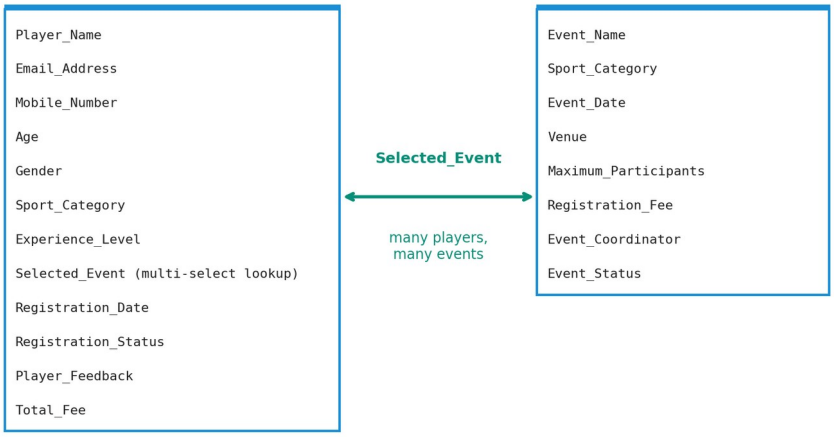


Figure 4.2: Entity–Relationship Diagram of the College Sports Administration System

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type
Player_Profile	Player_Name, Email_Address, Age, Sport_Category, Experience_Level, Selected_Event, Registration_Status, Total_Fee	Sports_Event	Many-to-Many (through the multi-select Selected_Event field)
Sports_Event	Event_Name, Sport_Category, Event_Date, Venue, Maximum_Participants, Registration_Fee, Event_Status	Player_Profile	Many-to-Many (events chosen by many players)

4.3 Form Design

Tables 4.2 and 4.3 give the field specification of the two forms as seen in the live application. Zoho Creator field types are named as they appear on the platform. The Player_Profile form has twelve fields and the Sports_Event form has eight fields, so the application has twenty fields of ten field types.

Table 4.2: Field Specification – Player_Profile Form

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Player_Name	Name	Yes	–	First name box
Email_Address	Email	Yes	Email format	Shown as a mail link in the report
Mobile_Number	Phone	Yes	Country code selector (+91)	Shown with the country code
Age	Number	Yes	Checked by Validate Age for Registration (12 or above)	Link name: age

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Gender	Drop Down	Yes	Fixed choices (for example Female, Other)	–
Sport_Category	Drop Down	Yes	Fixed choices (for example Cricket, Badminton)	Colour-coded in the report
Experience_Level	Drop Down	Yes	Beginner, Intermediate, Advanced	–
Selected_Event	Lookup (multi-select)	Yes	Choices come from the Sports_Event form	One or more events per player
Registration_Date	Date	Yes	dd-MMM-yyyy	–
Registration_Status	Drop Down	No	Approved, Rejected, Pending	Set to Pending by workflow
Player_Feedback	Multi Line	No	–	–
Total_Fee	Currency	Yes	Two decimal places	Update triggers a notification

Table 4.3: Field Specification – Sports_Event Form

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Event_Name	Single Line	Yes	–	–
Sport_Category	Drop Down	Yes	Fixed choices (for example Cricket, Basketball, Volleyball, Badminton)	Colour-coded in the report
Event_Date	Date	Yes	dd-MMM-yyyy	–
Venue	Single Line	Yes	–	–
Maximum_Participants	Number	Yes	–	–
Registration_Fee	Currency	Yes	Two decimal places	Shown with the $ symbol
Event_Coordinator	Email	No	Email format	–
Event_Status	Drop Down	Yes	Upcoming, Completed, Cancelled	–

4.4 Process and Workflow Design

Figure 4.3 shows the process that the application supports. The coordinator creates the sports events. A player submits a player profile; the Validate Age for Registration workflow checks the age and cancels the submission when the age is below 12. When the profile is saved, the Auto-Set Registration Status workflow sets the status to Pending. The coordinator then reviews the registration and sets the status to Approved or Rejected. When the total fee of a profile is

updated, the Notify on Fee Payment workflow checks the e-mail address of the player and sends the notification.

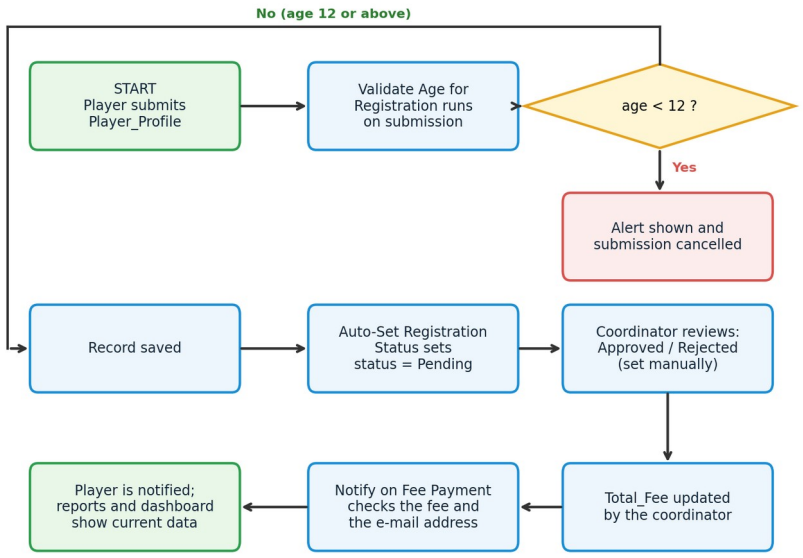


Figure 4.3: Workflow / Process Flow of the College Sports Administration System

4.5 User Interface and Navigation Design

The navigation has three entries in the top menu, as shown in Figure 4.4: the Dashboard, Player Profiles and Sports Events. The two form entries each contain the data-entry form and its report, so a user finds both in one place, and the Dashboard summarises both. The forms use left-aligned labels in a single column and the Submit and Reset buttons are at the end of the form.

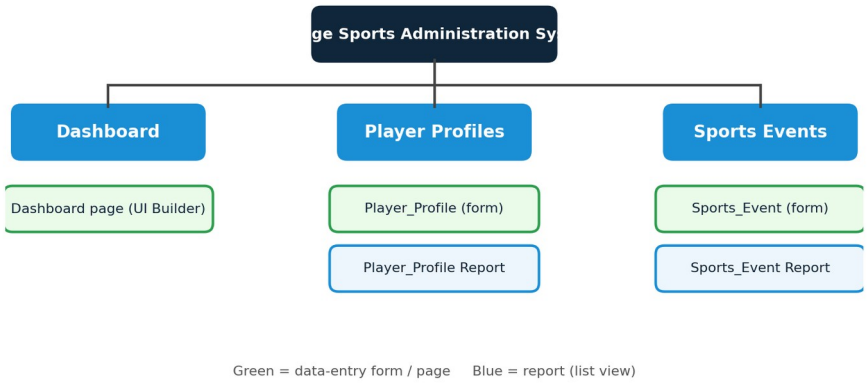


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Table 4.4 defines the permission matrix that the application is designed to enforce. A = Add, E = Edit, V = View, D = Delete, – = No access.

Table 4.4: Role–Permission Matrix

Component	Administrator	Sports Coordinator	Player
Form: Player_Profile	A E V D	A E V	A
Form: Sports_Event	A E V D	A E V	–
Report: Player_Profile Report	V	V	V (own)
Report: Sports_Event Report	V	V	V
Page: Dashboard	V	V	–

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “College Sports Administration System” was created in a Zoho Creator workspace and is opened from the link given in the Project at a Glance table. The top menu has three entries – Dashboard, Player Profiles and Sports Events – as seen in Figures 5.1 to 5.6. Dates use the dd-MMM-yyyy format. Table 5.1 summarises the configuration.

Table 5.1: Application Configuration Summary

Application name	College Sports Administration System
Link name	college-sports-administration-system
Method of creation	From scratch
Date format	dd-MMM-yyyy
Application menu	Dashboard (page), Player Profiles (form + report), Sports Events (form + report)
Form label placement	Left
Preview modes	Desktop, tablet and mobile previews are available from the application toolbar

5.2 Forms and Data Capture

Two forms capture the data of the application. The field specifications are given in Tables 4.2 and 4.3. Ten distinct field types are used: Name, Single Line, Multi Line, Email, Phone, Number, Drop Down, Lookup, Date and Currency.

Player_Profile form. The form in Figures 5.1 and 5.2 collects the player name, the e-mail address, the mobile number, the age, the gender, the sport category, the experience level, the selected events, the registration date, the registration status, the player feedback and the total fee. The mobile number field offers a country-code selector with +91 pre-selected, the age field shows a numeric placeholder, the Selected Event field lists the events of the Sports_Event form, and the total fee field shows a currency placeholder with the $ symbol. The Submit and Reset buttons are at the end of the form.

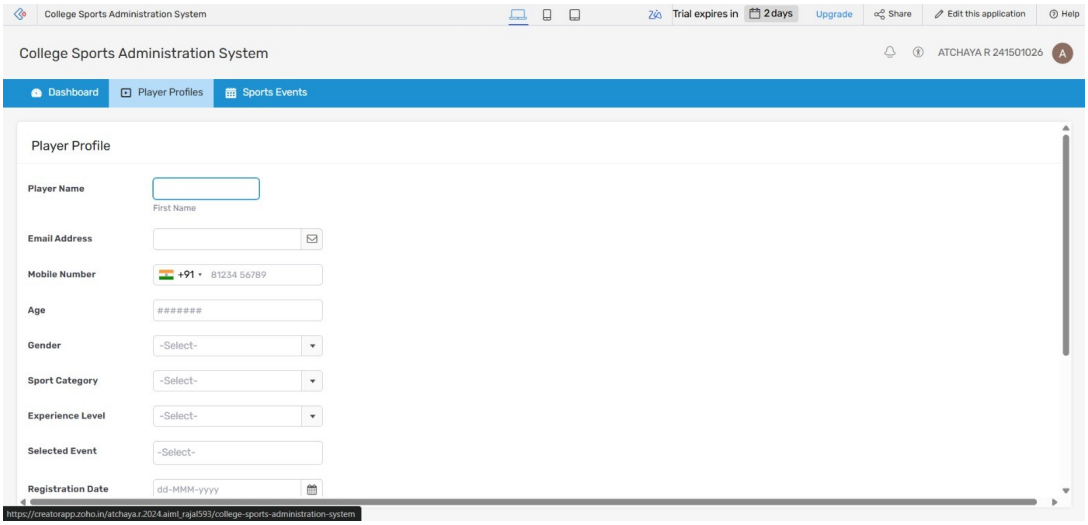


Figure 5.1: Player_Profile Form – Player Details, Category, Experience Level and Selected Event

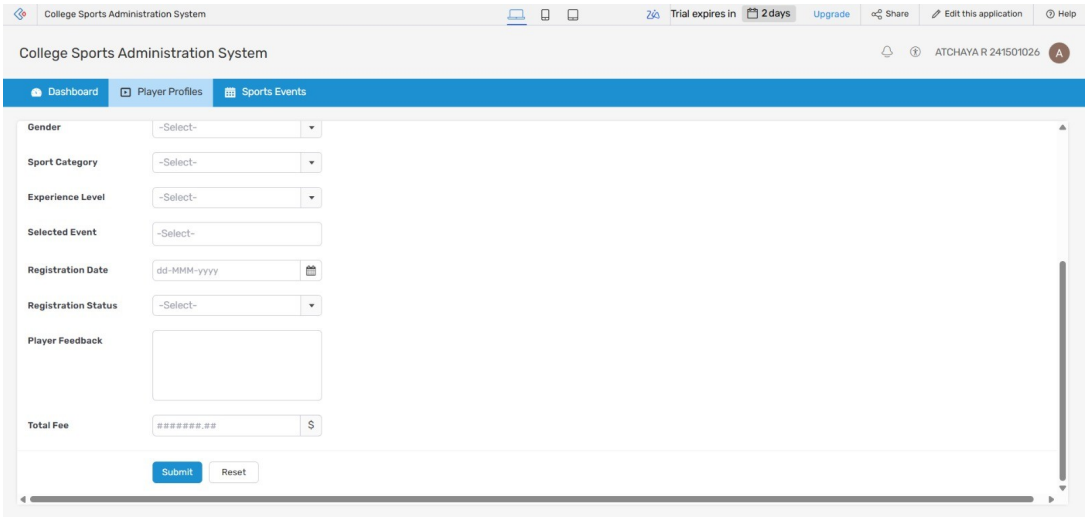


Figure 5.2: Player_Profile Form – Registration Status, Player Feedback, Total Fee and Submit Button

Sports_Event form. The form in Figure 5.3 records the event name, the sport category, the event date, the venue, the maximum participants, the registration fee, the event coordinator and the event status. The date field shows the expected dd-MMM-yyyy format, the registration fee field carries the $ symbol, and the event coordinator field is an e-mail field with a mail icon.

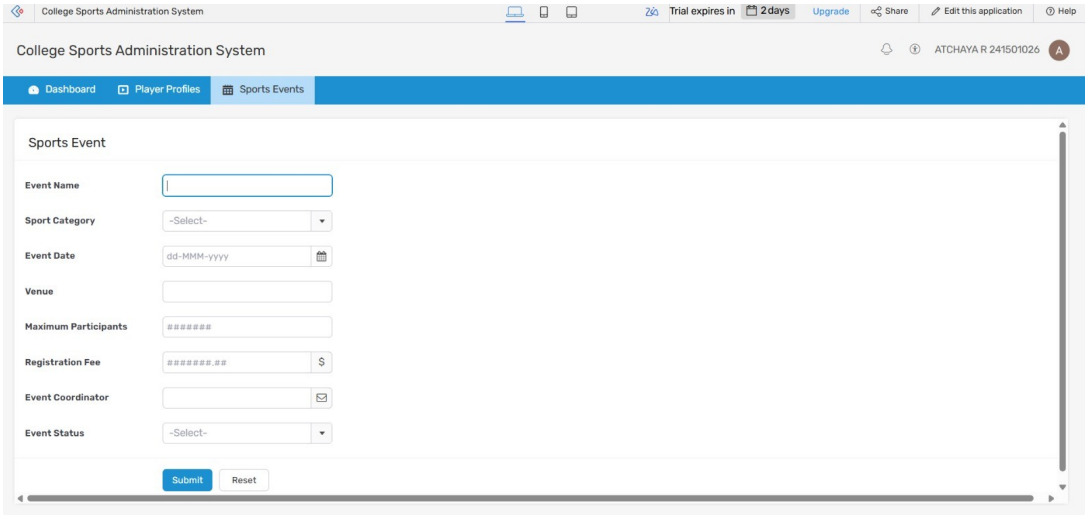


Figure 5.3: Sports_Event Form

5.3 Reports

Two list reports are available, as listed in Table 5.2. Each report shows the records of one form, has a search icon, an add button and a menu at the top, and displays the number of records at the bottom (“Showing 5 of 5”).

Table 5.2: Reports Implemented

Report Name	Type	Source Form	Columns Shown	Roles
Player_Profile Report (All Player Profiles)	List	Player_Profile	Player_Name, Email_Address, Mobile_Number, Age, Gender, Sport_Category, Experience_Level, Selected_Event, Registration_Date, Registration_Status	All with access
Sports_Event Report (All Sports Events)	List	Sports_Event	Event_Name, Sport_Category, Event_Date, Venue, Maximum_Participants, Registration_Fee, Event_Coordinator, Event_Status	All with access

Player_Profile Report. Figure 5.4 lists the five player profiles: David Kim, Emily Taylor, Marcus Johnson, Sarah Chen and Alex Rivera. For each player the report shows the e-mail address (as a mail link), the mobile number, the age, the gender, the colour-coded sport category, the experience level, the events chosen, the registration date and the registration status. Emily Taylor, for example, is a 23-year-old intermediate badminton player who has chosen four events and whose registration is Approved, while the registration of Sarah Chen is Pending.

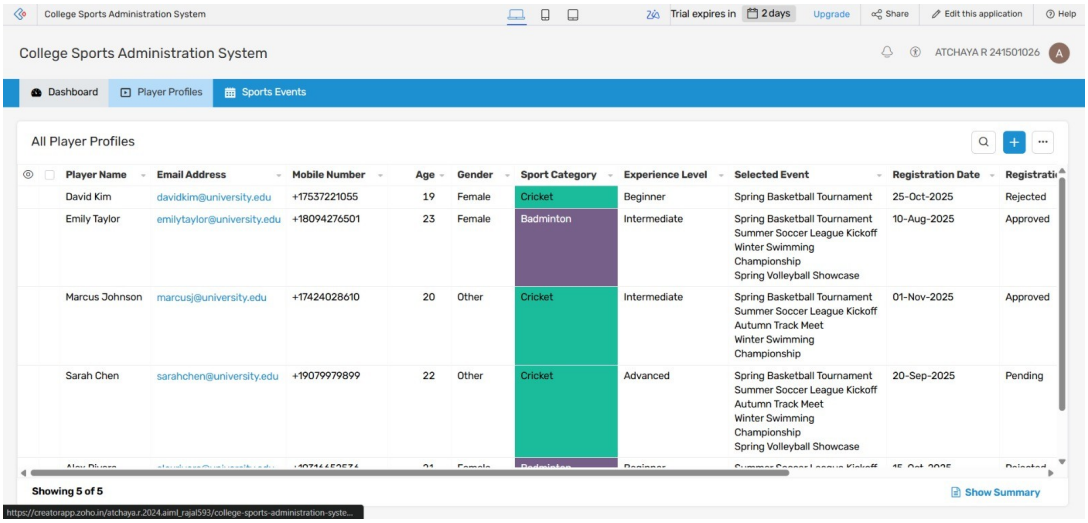


Figure 5.4: Player_Profile Report

Sports_Event Report. Figure 5.5 lists the five sports events: Spring Volleyball Showcase, Winter Swimming Championship, Autumn Track Meet, Summer Soccer League Kickoff and Spring Basketball Tournament. The report shows the colour-coded sport category, the event date, the venue, the maximum participants (from 150 to 1,200), the registration fee ($15.00 to $40.00), the event coordinator and the event status (Upcoming, Completed or Cancelled).

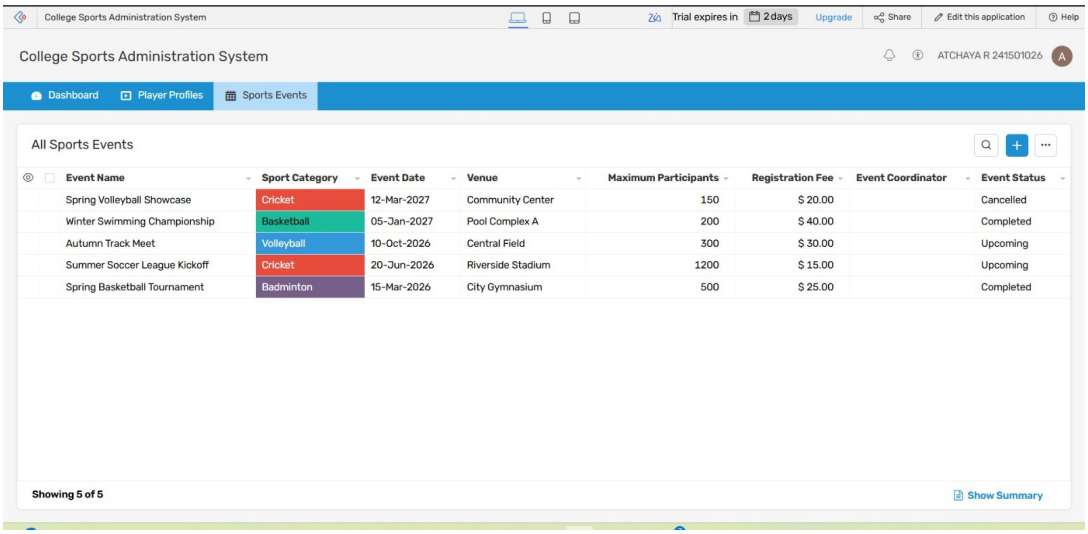


Figure 5.5: Sports_Event Report

5.4 Pages and Dashboards

A dashboard page was built in the UI Builder and is the first entry of the application menu. Figure 5.6 shows it. The top row has four coloured summary panels: the total of the total fees ($25,400.00), the total registration fees ($130.00), the number of player profiles (5) and the number of sports events (5). The Registration Status panel counts the profiles by status: two Approved, two Rejected, one Pending and five in total. The Quick Links panel opens the forms and reports with one click: Add New Player Profile, Show All Player Profiles, Add New Sports Event and Show All Sports Events. Two charts follow: a scatter chart of the registration fee against the maximum participants of the events, and a bar chart of the total fee by experience

level (Intermediate, Beginner and Advanced). The All Player Profiles table is shown at the bottom of the page.

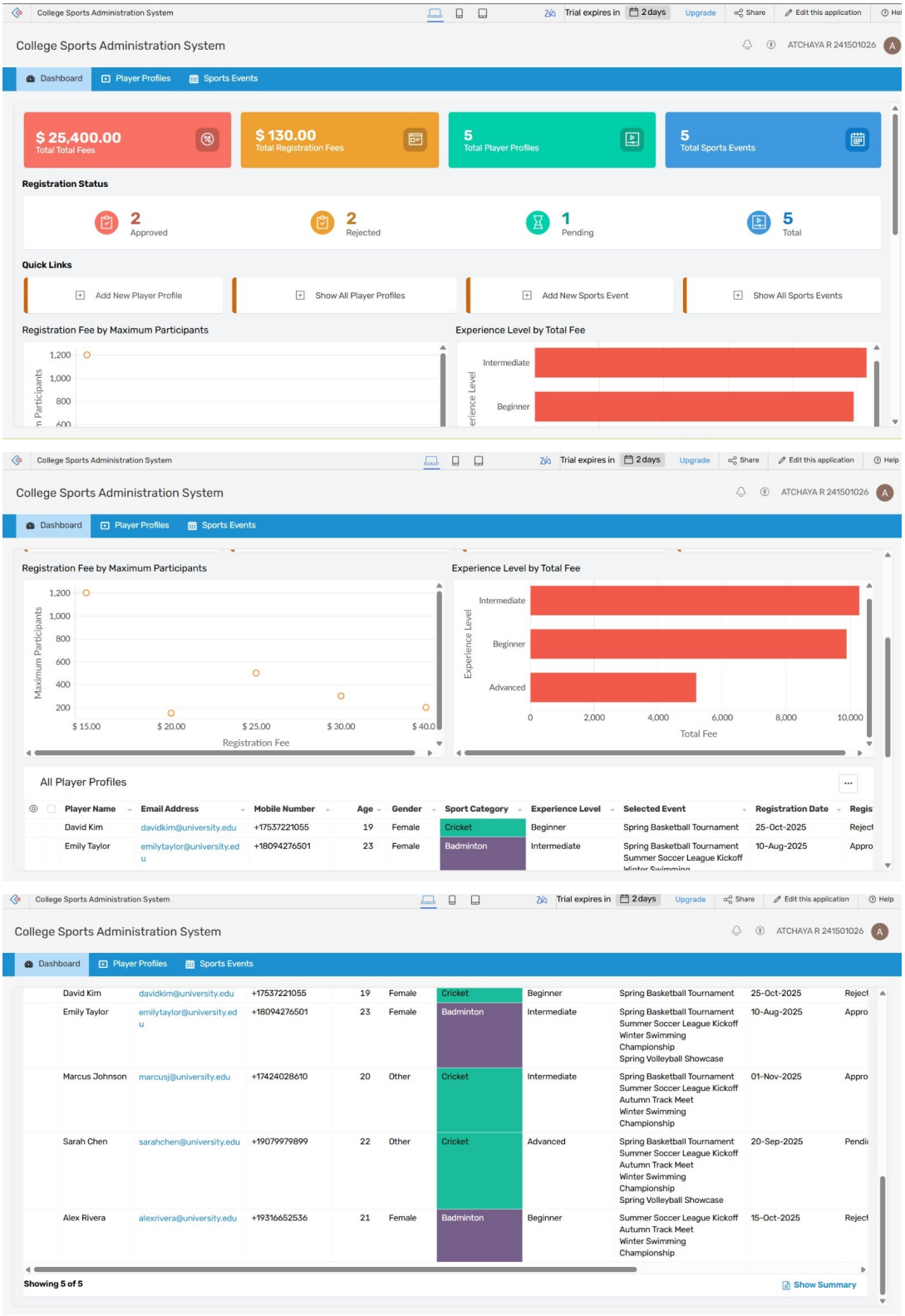


Figure 5.6: Dashboard – (a) Summary Panels, Registration Status and Quick Links, (b) Charts and Player Profiles, (c) Player Profiles Table

5.5 Deluge Scripting

The rules of the application are written in Deluge inside form workflows. Table 5.3 lists the workflow actions. The script of Auto-Set Registration Status is visible in Figure 5.7 and is reproduced in Code 5.1. It runs without a condition after a successful submission of a new Player_Profile record and assigns the value “Pending” to the registration_status field; an info statement writes the action to the log for debugging. The script of Notify on Fee Payment is visible in Figure 5.8. It runs when the Total Fee value is updated: it fetches the stored player profile record by its ID, compares the new total_fee with the stored value and, when the fee has really changed, reads the e-mail address of the player and checks that it is neither null nor empty before the notification is prepared. The script of Validate Age for Registration is visible in Figure 5.9 and is reproduced in Code 5.2. It runs on validation when a profile is created or edited; if the age is less than 12 it shows the alert “Player must be at least 12 years old to register.” and cancels the submission.

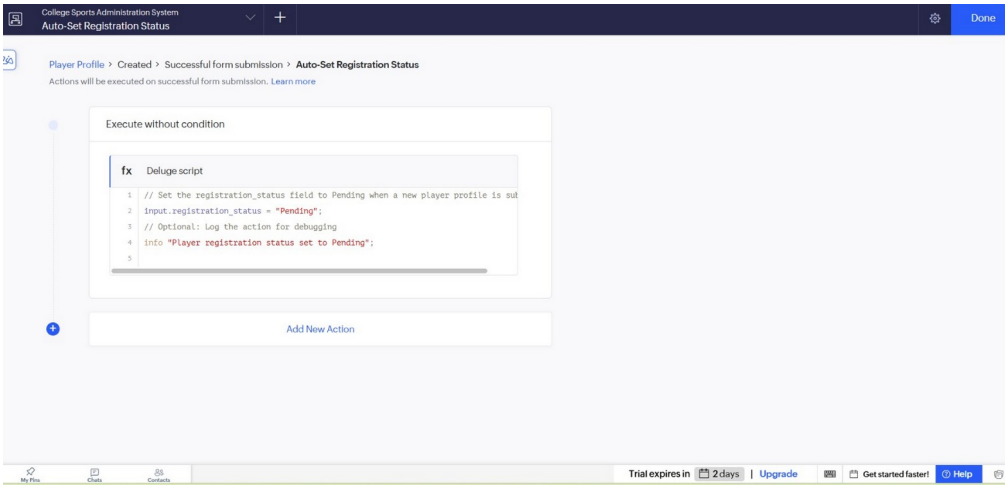


Figure 5.7: Deluge Script of the Workflow Auto-Set Registration Status

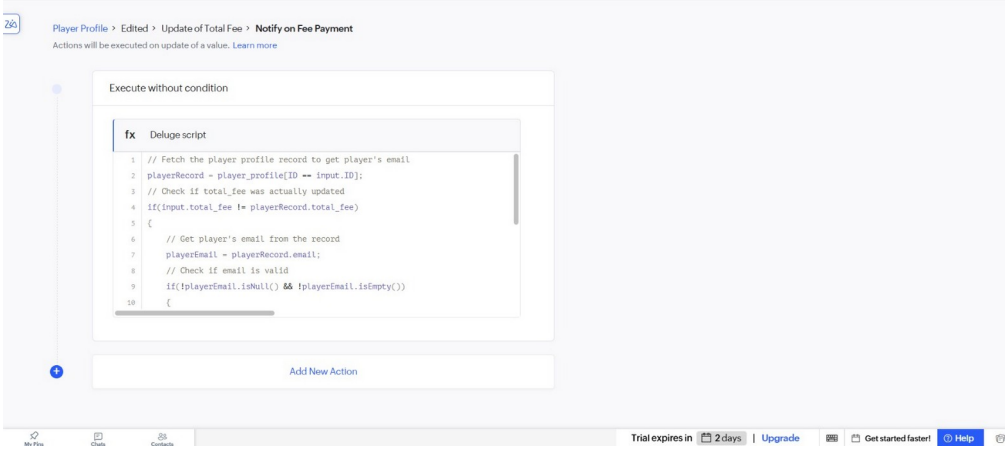


Figure 5.8: Deluge Script of the Workflow Notify on Fee Payment

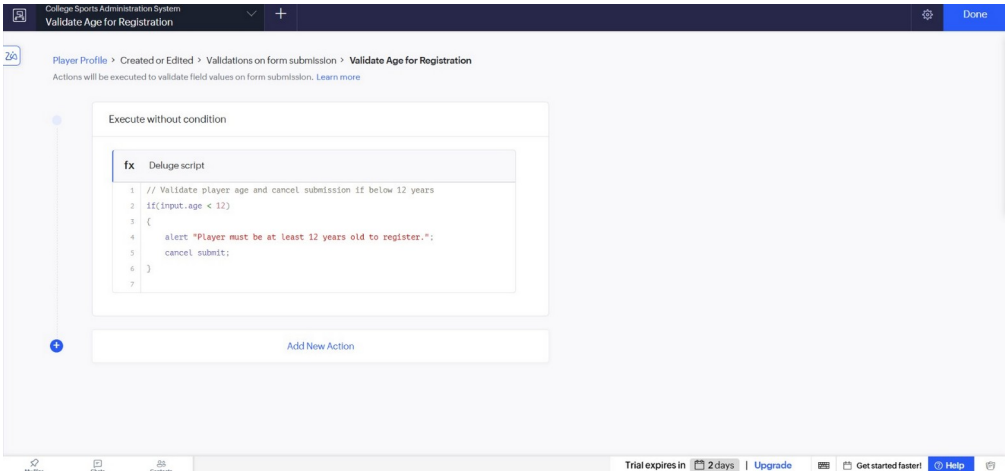


Figure 5.9: Deluge Script of the Workflow Validate Age for Registration

Table 5.3: Workflow Actions and Deluge Scripts

Workflow / Script	Form and Trigger	Purpose	Concepts Used
Auto-Set Registration Status	Player_Profile – created, successful form submission	Set the registration status of a new profile to Pending	Field assignment, comments, info statement
Notify on Fee Payment	Player_Profile – edited, update of Total Fee	Notify the player when the total fee is updated	Record fetch, comparison operators, logical operators, isNull / isEmpty
Validate Age for Registration	Player_Profile – created or edited, validations on form submission	Verify the age of the player before the record is saved	if condition, comparison operator, alert, cancel submit

The code of each script is given in Appendix C.

Code 5.1: Script of Auto-Set Registration Status (as visible in Figure 5.7)

```
// Set the registration_status field to Pending when a new player profile is
submitted
input.registration_status = "Pending";
// Optional: Log the action for debugging
info "Player registration status set to Pending";
```

Code 5.2: Script of Validate Age for Registration (as visible in Figure 5.9)

```
// Validate player age and cancel submission if below 12 years
if(input.age < 12)
{
    alert "Player must be at least 12 years old to register.";
    cancel submit;
}
```

5.6 Workflows and Automation

Three form workflows are configured and all of them are enabled, as shown in Figure 5.10. Each has one action and was created on 20-Oct-2026. All three belong to the Player_Profile form: one runs on successful submission of a new record, one runs when the Total Fee value is

updated in an edited record, and one runs on validations on form submission when a record is created or edited. The Sports_Event form is listed under “Forms without workflows”. No schedule, approval, Blueprint or batch workflow is configured. Table 5.4 lists the workflows.

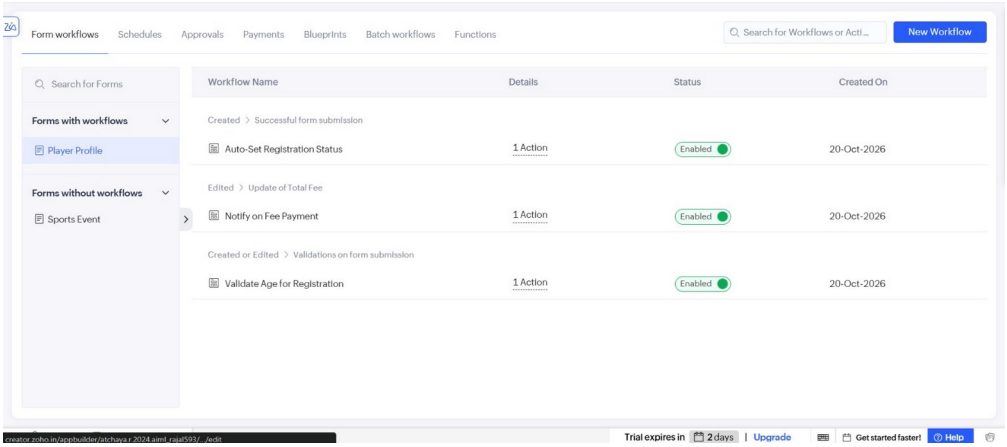


Figure 5.10: Form Workflows of the Player_Profile Form

Table 5.4: Workflows Implemented

Workflow Name	Form	Type / Trigger	Actions	Status
Auto-Set Registration Status	Player_Profile	Form workflow – created, successful form submission	1	Enabled
Notify on Fee Payment	Player_Profile	Form workflow – edited, update of Total Fee	1	Enabled
Validate Age for Registration	Player_Profile	Form workflow – created or edited, validations on form submission	1	Enabled

5.7 Integrations

No integration with a Zoho service or an external API is implemented in version 1.0, because the process is complete within the two forms. Table 5.5 lists the integrations that are planned.

Table 5.5: Integrations Planned for Future Versions (Not Yet Implemented)

Integration	Service / Endpoint	Method	Authentication	Purpose
Calendar	Zoho Calendar	Deluge task	Platform account	Add the event date as a reminder for the coordinator and the players
Analytics	Zoho Analytics	Built-in integration	Platform account	Advanced charts for fees, players and events
Payment	Payment gateway	REST API / connection	OAuth / API key	Collect the registration fee online

5.8 User Management and Role-Based Access

Access to the application is governed by the permission profiles of Zoho Creator. The application uses the default profiles of the platform (Administrator, Write, Read, Developer and the Customer portal profile). Record-level criteria, for example showing a player only the profile that the player has added, are not configured in version 1.0, so a user with access to a report sees all its records.

5.9 AI and Analytics Features

No Zia feature and no Zoho Analytics component is used in the application. The dashboard of Section 5.4 provides the summary panels and charts built with the UI Builder. Generative AI was used outside the application, to help draft the text of this report; all content was checked against the live application before inclusion.

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst function, JavaScript widget or REST API client) is used. The requirements of the application are met by forms, reports, a dashboard and short Deluge workflows, so pro-code would add maintenance effort without benefit. It would become justified for online payment of the registration fee, for generating participation certificates, or for integrations with other college systems.

5.11 Mobile and Portal Access

Zoho Creator applications are available on desktop, tablet and mobile screens, and the application toolbar provides previews for all three sizes. The application was not tested on the Zoho Creator mobile app and no customer portal has been configured; both are left for the next version, so no mobile or portal screenshot is included.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out at four levels. Form testing checked the field types of the two forms: the e-mail and phone formats, the date format, the numeric and currency fields and the lookup of events. Workflow testing checked the three form workflows by submitting profiles that should and should not pass the age validation and by watching the registration status and the fee notification. Report testing compared the records shown in the two reports and on the dashboard with the records entered. Acceptance testing is recorded in Section 6.5.

6.2 Test Cases

Table 6.1 lists 17 test cases that cover the functional requirements of Table 3.2 and include negative and boundary tests. The cases whose results come from the screenshots of the application carry the figure number; the other results follow the logic of the Deluge scripts shown in Section 5.5.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit a player profile with valid data	Fill all fields; Submit	Record saved	Five profiles listed (Figure 5.4)	Pass
TC-02	FR-01	Enter an invalid e-mail	Type “abc” in Email_Address; Submit	Validation error	Form rejects the invalid e-mail format	Pass
TC-03	FR-01	Mobile number with country code	Select +91 and type a number	Number stored with country code	+91 pre-selected (Figure 5.1); numbers shown with the country code (Figure 5.4)	Pass
TC-04	FR-01	Choose several events	Select more than one event in Selected_Event	All chosen events stored	Players with up to five events listed (Figure 5.4)	Pass
TC-05	FR-02	Add a sports event	Fill the event form; Submit	Record saved	Five events listed (Figure 5.5)	Pass
TC-06	FR-08	Date format	View the event dates and registration dates	dd-MMM-yyyy	12-Mar-2027 and 25-Oct-2025 (Figures 5.4, 5.5)	Pass
TC-07	FR-08	Fee format	View the registration fee	Currency with two decimals	$ 20.00 and $ 40.00 (Figure 5.5)	Pass
TC-08	FR-03	Status of a new profile	Submit a new player profile	Registration status = Pending	Auto-Set Registration Status assigns “Pending” (Figure 5.7)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-09	FR-04	Age below 12	Enter age 11; Submit	Alert shown; record not saved	Alert “Player must be at least 12 years old to register.” and submission cancelled (Figure 5.9)	Pass
TC-10	FR-04	Boundary age	Enter age 12; Submit	Record saved	Condition age < 12 is false, so the record is saved	Pass
TC-11	FR-04	Age checked on edit	Edit a profile and set age 10	Alert shown; change not saved	Workflow runs when a record is created or edited (Figure 5.10)	Pass
TC-12	FR-05	Fee changed	Change Total_Fee of a profile with a valid e-mail	Player is notified	Script compares the fee and checks the e-mail (Figure 5.8)	To verify
TC-13	FR-05	Fee not changed	Edit a profile without changing Total_Fee	No notification	Condition total_fee != stored value is false (Figure 5.8)	Pass
TC-14	FR-09	Dashboard totals	Open the Dashboard	Totals match the records	$25,400.00, $130.00, 5 profiles, 5 events; 2 approved, 2 rejected, 1 pending (Figure 5.6)	Pass
TC-15	FR-07	Search a record	Use the search icon in a report	Matching record shown	Search is available in both reports (Figures 5.4, 5.5)	Pass
TC-16	FR-06	Workflows are enabled	Open the Workflow page	All workflows enabled	All three show Enabled (Figure 5.10)	Pass
TC-17	NFR-03	Layout preview	Use the mobile and tablet previews	Layout adapts	Previews are offered in the toolbar	Pass

6.3 Debugging Deluge Scripts

The script of Validate Age for Registration (Code 5.2) was debugged by reasoning through three cases with sample values: an age below 12, an age of exactly 12 and an age above 12. Because the condition uses “less than 12”, the age of 12 is accepted, which matches the rule that a player must be at least 12 years old. The Auto-Set Registration Status script writes an info statement, so the execution log shows that the status was set to Pending. The Notify on Fee Payment script compares the new fee with the stored record, so an edit that does not change the fee does not trigger a notification.

Debugging a Deluge workflow on Zoho Creator uses the following techniques: info statements to print the values of fields, the error message that the platform shows when a script fails on submission, and the execution history of the workflow. Each workflow should be tested with a

valid record, with a record that breaks the rule, and with missing values so that null values do not break the script; the isNull and isEmpty checks of the notification script serve this purpose.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Planned	Status
BUG01	A player can choose events of any sport; for example a cricket player is registered for a basketball tournament (Figure 5.4)	Selected_Event lists all events and is not filtered by the sport category of the player	Filter the Selected_Event lookup by Sport_Category, or validate the choice in a workflow	Open
BUG02	The event status is not in step with the event date; an event dated in 2027 is marked Completed while an event dated 10-Oct-2026 is still Upcoming (Figure 5.5)	Event_Status is chosen by hand and no workflow updates it	Add a scheduled workflow that sets Upcoming, Completed from the Event_Date	Open
BUG03	Total_Fee is typed by hand and does not follow the registration fees of the chosen events	No calculation links Total_Fee to Registration_Fee	Calculate Total_Fee from the selected events in a workflow	Open

6.5 User Acceptance Testing

Table 6.3 records the feedback of users who tried the application in their assigned role.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Atchaya R	Developer/Tester	The application is easy to use. Players, events, registration status and fees can be managed effectively, and the dashboard gives a quick summary.	Verified the forms, tested the workflows and corrected validation errors.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator separates an application into Development, Stage and Production environments so that changes can be tested before users see them. Version 1.0 of this application was built and tested in the Development environment. The Stage and Production environments depend on the edition of the account; the account used for this project was a trial account (the header of the screenshots shows two days of the trial remaining). A staged deployment will therefore be performed when the application is moved to a paid edition, and the rollback plan is to re-import an exported copy of the version 1.0 definition. Table 7.1 gives the deployment checklist with the present status.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)	Remarks
1	All test cases executed in Development / Stage	Partial	16 of 17 passed; TC-12 to be verified
2	Sample/test data removed or isolated	✗	Sample records still present
3	Roles, permissions and users configured in Production	Partial	Default profiles only
4	Connections authorised in Production	N/A	No connections used
5	Schedules and workflows enabled	✓	3 form workflows enabled; no schedules
6	Application published and URL verified	✓	Link given in Project at a Glance
7	Version / release notes recorded	✓	Export the application definition and note it

7.2 Performance Optimisation

The data volume of version 1.0 is small, so no measurable performance problem arose and no before-and-after timing was taken. The design nevertheless applies measures that keep the application efficient as it grows, listed in Table 7.2.

Table 7.2: Performance Optimisation Measures

Optimisation Applied	Component	Benefit	Measured?
The age and status scripts use only the values of the submitted record and no database fetch	Validate Age for Registration, Auto-Set Registration Status	No extra database calls per submission	No
The notification script fetches a record only when a fee value is updated	Notify on Fee Payment	Fetch happens only on an edit of the fee	No

Optimisation Applied	Component	Benefit	Measured?
Drop-downs for category, gender, level and status instead of free text	Both forms	Consistent values; easier filtering	No
Each report shows only the key fields of its form	Reports	Faster listing on small screens	No

7.3 Security Features

Security rests on the controls of the platform and on the configuration of the application, as summarised in Table 7.3. Record-level access criteria, two-factor authentication and IP restrictions are not configured in version 1.0 and are recommended before the application is used with real player data, because the forms hold e-mail addresses and mobile numbers.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Field-type validation	Email, Phone, Date, Number, Currency and Lookup fields	Malformed or invalid input
Validation workflow	Validate Age for Registration	Under-age players entering the database
E-mail checks in script	Notify on Fee Payment (isNull, isEmpty)	Notifications to empty or invalid addresses
Permission profiles	Default Zoho Creator profiles	Unauthorised access or modification
Secure access channel	Platform access over HTTPS	Interception of data in transit

7.4 Governance and Administration

Field names follow one convention (words joined by underscores, such as Maximum_Participants), and workflow names describe their purpose (for example Validate Age for Registration). The date format is fixed at application level so that all users see the same format. The audit and change-history facilities of the platform are available to the administrator, subject to the edition in use. The Administrator is responsible for adding and removing users, for reviewing the permission profiles and for licence management. Two items need attention: the account must be upgraded from the trial before the application is used for a real sports season, and an exported copy of the application definition should be kept as a backup and version record.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application was walked through with one realistic scenario: the coordinator creates five sports events, five players register through the player profile and choose their events, the registration status is set to Pending, the coordinator approves or rejects the registrations and updates the fees, and the coordinator reviews the dashboard and the reports. Table 8.1 lists the steps and the figures that show the evidence. At the end of the scenario the application holds five sports events with registration fees from $15.00 to $40.00 and five player profiles, of which two are Approved, two are Rejected and one is Pending, with total fees of $25,400.00 shown on the dashboard.

Table 8.1: Scenario Walkthrough and Evidence

Step	Action	Role	Evidence
1	The coordinator creates a sports event with its category, date, venue, participants and fee	Sports Coordinator	Figures 5.3 and 5.5
2	The player opens the profile form and enters the details, the category and the events	Player	Figures 5.1 and 5.2
3	The age is validated and the record is saved with the status Pending	System	Figures 5.7, 5.9 and 5.10
4	The coordinator reviews the profiles and sets Approved or Rejected	Sports Coordinator	Figure 5.4
5	The coordinator updates the total fee and the player is notified	Sports Coordinator / System	Figure 5.8
6	The coordinator reads the totals, the status counts and the charts of the dashboard	Sports Coordinator	Figure 5.6

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Forms for events and players linked by the selected events	Sections 4.3, 5.2	2 forms, 20 fields of 10 field types; Selected_Event links players to events
2. Automatic status and age validation	Sections 5.5, 5.6	Workflows Auto-Set Registration Status and Validate Age for Registration enabled
3. Fee notification	Sections 5.5, 5.6	Workflow Notify on Fee Payment enabled
4. Reports and dashboard for the coordinator	Sections 5.3, 5.4	2 list reports with search and 1 dashboard with summary panels, status counts and charts; filters and grouping pending

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
5. Testing, defects and improvements	Chapter 6, Section 9.2	17 test cases, 3 defects logged, 7 enhancements listed

8.3 Comparison with the Existing System

Table 8.3: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording a player	Paper, message or e-mail, then copied to a sheet	One form with validated fields	Complete and uniform records
Age checking	By hand	Validate Age for Registration on submission	No under-age registrations
Event choice	Noted in a register	Selected_Event lookup linked to the Sports_Event form	Player and events in one record
Registration status	By hand	Set to Pending by a workflow	Always starts consistent
Fee information	Told to the player in person	E-mail notification when Total_Fee is updated	Player informed
Report generation	Compiled by hand	Two reports and a dashboard available at any time	Immediate visibility
Data accessibility	One file with one person	Cloud access from desktop, tablet and mobile	Shared, controlled access

8.4 Limitations of the Low-Code Approach

Several limitations were met during the work. First, the Selected_Event lookup is not filtered by the sport category, so a player can choose events of another sport (BUG-01). Second, the event status and the total fee are not calculated from the event date and the registration fees, so they can differ from the real situation (BUG-02 and BUG-03). Third, the reports have no filters or grouping, so the coordinator cannot yet list only the pending registrations. Fourth, features such as an approval process, scheduled status updates, certificates and online payment need approval or schedule workflows or integrations that are not yet built. Fifth, the edition limits what can be used: the application runs on a trial account, and staged environments depend on the edition. Finally, the application is specific to Zoho Creator, so moving to another platform means rebuilding it. The workaround adopted was to keep the data model small and to document the rules in plain language.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The forms, fields, reports, dashboard and navigation were created by configuration, which is the no-code end of the spectrum. The three form workflows, especially the age validation and the fee notification, are the low-code layer: a few lines of Deluge replace a manual check and a

manual message. The pro-code layer would be needed for online payment, for participation certificates and for integrations with other college systems. An application should move to pro-code only when Deluge cannot meet a requirement, because each step to the right increases the skill needed to maintain it.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The College Sports Administration System was built on Zoho Creator to replace the notices, registers and spreadsheets used to manage sports events and player registrations. It has two forms – Player_Profile and Sports_Event – linked through the Selected_Event field, with 20 fields of 10 field types, two list reports, a dashboard page and three enabled form workflows. Drop-downs and a validation workflow keep the data clean, a workflow sets the registration status of every new profile to Pending, and a notification workflow informs the player when the total fee is updated.

Testing showed that data capture, the age validation, the default status, the reports and the dashboard work as designed, and it exposed three defects: the events are not filtered by sport, the event status does not follow the event date and the total fee is not calculated. The project shows that a working administration application can be assembled quickly on a low-code platform, and that its rules can be automated with small scripts.

9.2 Future Enhancements

- Filter the Selected_Event lookup by the sport category of the player.
- Add a scheduled workflow that sets the event status from the event date.
- Calculate the total fee from the registration fees of the selected events.
- Add an approval process so that the coordinator can approve or reject a registration and the status is recorded automatically.
- Add filters and grouping to the reports, such as pending registrations or upcoming events.
- Collect the registration fee online through a payment gateway and issue participation certificates.
- Configure a player portal with record-level criteria so that each player sees only their own profile, and upgrade from the trial edition to use the Development, Stage and Production environments.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in form, report and dashboard design, Deluge workflows, validation, notification, testing and defect tracking, and in recognising where a low-code platform needs scripting. These skills correspond to the roles of citizen developer, low-code developer and solutions consultant [7]. The project also showed that a low-code solution is judged by how well it fits the real process, and not only by how quickly it was built.

REFERENCES

[1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: 21 October 2026).

[3] D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi, and M. Wimmer, “Low-code development and model-driven engineering: Two sides of the same coin?,” Software and Systems Modeling, vol. 21, no. 2, pp. 437–446, 2022, doi: 10.1007/s10270-021-00970-2.

[4] R. Sanchis, Ó. García-Perales, F. Fraile, and R. Poler, “Low-code as enabler of digital transformation in manufacturing industry,” Applied Sciences, vol. 10, no. 1, Art. no. 12, 2020, doi: 10.3390/app10010012.

[5] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/ (Accessed: 21 October 2026).

[6] A. Sahay, A. Indamutsa, D. Di Ruscio, and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA), 2020, pp. 171–178, doi: 10.1109/SEAA51224.2020.00036.

[7] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/ (Accessed: 21 October 2026).

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = fully implemented, Partial = implemented in part, N = not implemented in version 1.0.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Menu: Dashboard, Player Profiles, Sports Events	5.1	Figs. 5.1–5.6
II	Forms with ≥ 8 distinct field types	Y	Both forms (10 types)	5.2	Figs. 5.1–5.3
II	Lookup field (relationship between forms)	Y	Player_Profile → Sports_Event (Selected_Event, multi-select)	4.2	Fig. 4.2, Fig. 5.1
II	Subform (parent–child data)	N	Not required in this design	4.2	–
II	Field validations / layout rules	Y	Field types, age validation workflow, workflow-set status	4.3, 5.6	Tables 4.2, 4.3
II	List report with filter, grouping, custom action	Partial	2 list reports with search; no filter or grouping	5.3	Table 5.2
II	Kanban / Calendar / Timeline / Map report	N	–	5.3	–
II	Pivot table / chart report	N	–	5.3	–
II	Page / dashboard built with UI Builder	Y	Dashboard with summary panels, status counts, quick links and charts	5.4	Fig. 5.6
II	Deluge – form events (validate / success / field actions)	Y	3 workflows on validation, success and field update	5.5, 5.6	Figs. 5.7–5.10
II	Deluge – functions, fetch, loops, collections	Partial	Record fetch and conditions; no functions or loops	5.5	Fig. 5.8
II	Email / in-app notifications	Y	Notify on Fee Payment	5.5	Fig. 5.8
II	Form workflow	Y	3 enabled form workflows	5.6	Table 5.4
II	Scheduled workflow	N	Planned (event status)	9.2	–
II	Approval process	N	Planned	9.2	–
II	Blueprint	N	–	5.6	–
II	Batch / report workflow	N	–	5.6	–
II	Testing & debugging evidence	Y	17 test cases, 3 defects	6	Tables 6.1, 6.2
III	Integration with a Zoho service	N	Planned (Zoho Calendar, Zoho Analytics)	5.7	Table 5.5

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Connection / REST API call (invokeurl)	N	Planned (payment gateway)	5.7	–
III	Webhook or Creator REST API	N	–	5.7	–
III	Users, roles and permission profiles	Partial	Default profiles	5.8	Table 4.4
III	Record-level access criteria / customer portal	N	Planned	5.8	–
III	Zia / AI feature	N	–	5.9	–
III	Analytics (charts / Zoho Analytics)	N	Dashboard charts built in UI Builder; Zoho Analytics planned	5.9	–
IV	Performance optimisation applied & measured	Partial	Design measures; not measured	7.2	Table 7.2
IV	Deployment across environments	N	Development only	7.1	Table 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Validation, e-mail checks, default profiles	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Not reviewed	7.4	–
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Justified in report	5.10	–
V	Low-code limitations & spectrum analysis	Y	Discussion in report	8.4, 8.5	–
–	Mobile app / portal access	N	Not tested	5.11	–

Total features fully implemented: 10 / 32 (partially implemented: 5)

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – COMPLETE DELUGE SOURCE CODE

This appendix presents the Deluge scripts used in the College Sports Administration System to set the registration status, to notify the player when the total fee is updated and to validate the age of the player. All three scripts belong to the Player_Profile form.

C.1 Auto-Set Registration Status – Player_Profile, Successful Form Submission (Created)

The following script sets the registration status of a new player profile to Pending.

```
// Set the registration_status field to Pending when a new player profile is
submitted
input.registration_status = "Pending";
// Optional: Log the action for debugging
info "Player registration status set to Pending";
```

C.2 Notify on Fee Payment – Player_Profile, Update of Total Fee (Edited)

The following lines of the script fetch the stored record, check that the total fee was updated and check the e-mail address of the player. The lines that follow in the editor (the notification to the player) are to be pasted here from the Deluge editor.

```
// Fetch the player profile record to get player's email
playerRecord = player_profile[ID == input.ID];
// Check if total_fee was actually updated
if(input.total_fee != playerRecord.total_fee)
{
    // Get player's email from the record
    playerEmail = playerRecord.email;
    // Check if email is valid
    if(!playerEmail.isNull() && !playerEmail.isEmpty())
    {
        // ... remaining lines to be pasted from the Deluge editor
    }
}
```

C.3 Validate Age for Registration – Player_Profile, Validations on Form Submission (Created or Edited)

The following script cancels the submission when the age of the player is below 12 years.

```
// Validate player age and cancel submission if below 12 years
if(input.age < 12)
{
    alert "Player must be at least 12 years old to register.";
    cancel submit;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the application link https://creatorapp.zoho.in/atchaya.r.2024.aiml_rajal593/college-sports-administration-system/#Section_1 in Google Chrome, Microsoft Edge or Mozilla Firefox.
2. Sign in with your Zoho account credentials. Your profile decides which forms and reports you can see.
3. On a mobile phone, open the same link in the browser or open the application in the Zoho Creator mobile app.
4. Use the top menu to move between the Dashboard, Player Profiles and Sports Events.

D.2 Instructions for the Player

1. Open Player Profiles and click Add (or open the Player_Profile form).
2. Enter the player name, e-mail address, mobile number, age and gender, and select the sport category and the experience level.
3. Select the events you want to take part in, enter the registration date and click Submit. If the age is below 12 a message is shown and the profile is not saved.
4. Open the Player Profiles report to check that the profile is listed with the registration status Pending.

D.3 Instructions for the Sports Coordinator

1. Open Sports Events and add each event with its sport category, date, venue, maximum participants, registration fee, event coordinator and event status.
2. Open the Dashboard to see the total fees, the number of players and events, the registration status counts and the charts.
3. Open the Player Profiles report to review each registration; edit a record to set the registration status to Approved or Rejected and to update the total fee. The player is notified of a fee update.
4. Use the search icon to find a player or an event, and click a record to read its details.

D.4 Instructions for the Administrator

1. Open Edit this application to change a form, a report, the dashboard or a workflow in the builder.
2. Open the Workflow page to check that the three form workflows are enabled.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	ATCHAYA R	**Register No.**	2116241501026
Project Title	College Sports Administration System	**Date**	21/10/2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	**Criterion**	**Indicators**	**Max. Marks**	**Marks Awarded**
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
TOTAL			**100**	

Table E.2: Performance Level Descriptors

Level	**% of Max. Marks**	**Descriptor**
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.

Level	% of Max. Marks	Descriptor
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Signature of External Examiner

Name: _____ *Name:* _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	ATCHAYA R	**Register No.**	2116241501026
Department / Section	AIML/A	**Date of Viva**	21/10/2026
Application URL Verified	Yes / No	**Demo Duration**	____ *m*

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ *Name:* _____